

RESCUE JUNIOR - Bill of Materials (BOM)

Team name: Anyang

Every Line/Maze team has to submit a bill of materials for their robot

Instructions:

*Enter Team name above
*All costs need to be in local currency and their approximate conversion into dollars.
*All components worth less can be summarized in one line quantity 1 and their total cost (e.g. Screws)
*In the Local Currency Name column, the team must enter the name of the local currency, if it is different from US dollars.
*In the software sheet, in the Author column, the name of the designer, writer, or company that owns the software or library should be entered. If the team created their own algorithm, dataset, or AI model, they can include their names as authors.
*On the hardware sheet, in the "Author" column, enter the name of the company, engineer, or part manufacturer. If the team made a custom part, such as the robot housing, the team should include their names as the authors.

IMPORTANT The "Hardware" sheet is not required for simulation

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#	Name	Software's Tool/Library	Description
1	Webots	Software's Tool	Open-source 3D robot simulator. Provides physics engine, sensor simulation (camera, lidar, GPS, distance sensors), and Python controller API used for the competition environment.
2	Python	Software's Tool	Primary programming language for robot controller logic. Used for all sensor processing, pathfinding, YOLO inference, mapping, and GUI code.
3	Visual Studio Code	Software's Tool	Source code editor used for writing and debugging the robot controller. Extensions used: Python, Pylance.
4	GitHub	Software's Tool	Version control and collaboration platform. Used to manage source code, track changes, and share code among team members throughout development.
5	RoboFlow	Software's Tool	Web-based platform for annotating training images and exporting YOLOv5-compatible datasets. Used to label victim images (H/S/U) for custom model training.
6	Google Colab	Software's Tool	Cloud-based Jupyter notebook environment with free GPU access. Used to train the YOLOv5 custom model on the annotated victim dataset.
7	YOLOv5	Library	Real-time object detection model. Fine-tuned to classify three victim types: Harmed (H), Stable (S), and Unharmed (U) from camera images during simulation.

8	PyTorch	Library	Deep learning framework used as the backend for loading and running the trained YOLOv5 model. Handles tensor operations and GPU/CPU inference.
9	OpenCV	Library	Computer vision library used for camera image preprocessing, color space conversion (BGR→HSV/RGB), contour detection, and victim color circle identification.
10	NumPy	Library	Numerical computing library. Used extensively for lidar point cloud processing, grid map array operations, coordinate transformations, and mathematical calculations.
11	Matplotlib	Library	Plotting library used to visualize the occupancy grid map and cloud map during development and debugging. Saved map images for analysis.
12	Tkinter / Pillow	Library	Tkinter is Python's built-in GUI library; Pillow (PIL) provides image handling. Together they power the real-time dashboard showing sensor data, map, and camera feeds during simulation.
13	scikit-image	Library	Image processing library. Used specifically for the skimage.draw.line function to rasterize lidar rays onto the 2D occupancy grid map for wall detection.
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Name of the Local Currency	Total Cost Local Currency
Brazilian Real	R\$66.24

Source	Autor	Currency)
https://cyberbotics.com/	Cyberbotics Ltd.	FREE
https://www.python.org/	Python Software Foundation	FREE
https://code.visualstudio.com/	Microsoft Corporation	FREE
https://github.com/	GitHub, Inc.	FREE
https://roboflow.com/	Roboflow, Inc.	FREE
https://colab.research.google.com/	Google Research	\$66.24
https://github.com/ultralytics/yolov5	Ultralytics LLC	FREE

[illegible]

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Total Cost (\$) U.S.A. Dollars
\$66.24

Cost (USA
FREE
FREE
FREE
FREE
FREE
\$66.24
FREE

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